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Articular System

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Course Outline

- Introduction
- Classification of Joints
- Diarthroses (Synovial Joints)

1. Introduction

Definition

A joint (articulation) is the junction between two or more skeletal components.

A joint connects either two bones (e.g., sacroiliac joint), two cartilages (e.g., larynx), or a bone and a cartilage (e.g., sternochondral joint).

The scientific study of joints is known as arthrology.

Functions

Joints fulfill two main primary roles:

- Guiding and enabling movement.
- Providing structural stability and consolidation to the skeletal components.

2. Classification of Joints

Joints are classified into three anatomical types based on the nature of the joining structural tissue:

Synarthroses

Also known as immovable or fibrous joints.

In synarthroses, the bony elements are united by fibrous connective tissue, which may undergo secondary ossification/calcification.

Examples: Distal tibiofibular joint, interosseous membranes, cranial and facial sutures, and the fusion of the three components of the hip bone (ilium, ischium, and pubis).

Amphiarthroses

Also known as slightly movable or cartilaginous joints.

In amphiarthroses, the bony elements are joined by cartilaginous tissue (forming a fibrocartilaginous block). This type of joint is typically located along the midline.

Examples: Intervertebral joints, pubic symphysis.

Diarthroses

Also known as freely movable or synovial joints.

They are characterized by their significant mobility and wide range of motion.

3. Diarthroses (Synovial Joints)

Anatomical Description

A diarthrodial joint consists of four principal anatomical structures:

1. Articular Surfaces

- Located at the epiphyses of long bones, or at the angles and surfaces of flat bones.
- Possess defined geometric shapes, covered with smooth articular (hyaline) cartilage.
- In cases of structural incongruity, adapting fibrocartilaginous structures interpose to ensure proper fit.

2. Joint Cavity

The fluid-filled space separating the opposing articular surfaces.

3. Retaining Structures (Means of Union)

Structures enclosing the articular surfaces, represented by three components:

- **Capsule (Articular Capsule):** A fibrous envelope enclosing the joint cavity.
- **Passive Ligaments:** Fibrous bands that reinforce the articular capsule.
- **Active Ligaments:** Tendons of muscles adjacent to the joint; they stabilize and actively mobilize the joint.

4. Synovial Membrane

The specialized gliding membrane of a diarthrodial joint. It lines the internal surface of the articular capsule and attaches at the margins of the articular cartilages.

The synovial membrane secretes a viscous fluid called synovial fluid (synovia), which serves a triple function:

- Resorption of metabolic waste products.
- Nutritional support for the avascular articular cartilage.
- Lubrication to reduce friction during movement.

Classification of Diarthroses

Diarthroses are classified into six structural types based on the shape of their articular surfaces:

1. Spheroid Joint

- **Configuration:** Articular surfaces are complementary spherical segments (a convex ball fitting into a concave socket).
- **Mobility:** Multiaxial (provides movement around three axes).
- **Example:** Glenohumeral (shoulder) joint.

2. Ellipsoid Joint

- **Configuration:** Articular surfaces are elliptical, consisting of a concave and a convex surface.
- **Mobility:** Biaxial (provides movement around two axes).
- **Example:** Radiocarpal (wrist) joint.

3. Ginglymus Joint

- **Configuration:** Articular surfaces are pulley-shaped.
- **Mobility:** Uniaxial (provides movement around a single axis).
- **Example:** Humeroulnar joint.

4. Cylindrical Joint

- **Configuration:** Articular surfaces are cylindrical (composed of a central bony pivot surrounded by a ring).
- **Mobility:** Uniaxial (provides rotational movement around a single axis).
- **Example:** Proximal radioulnar joint.

5. Plane Joint

- **Configuration:** Articular surfaces are flat or nearly flat.
- **Mobility:** Multiaxial, permitting limited gliding movements in multiple directions.
- **Example:** Acromioclavicular joint.

6. Saddle Joint (Sellar Joint)

- **Configuration:** Articular surfaces are opposingly concave and concave, fitting together like 'a rider on a saddle'.
- **Mobility:** Biaxial (provides movement around two axes).
- **Example:** Carpometacarpal joint of the thumb (trapeziometacarpal joint).

Adapting Structures

Intra-articular fibrocartilaginous structures interposed between non-congruent articular surfaces to ensure optimal alignment and congruence. There are three types:

1. Meniscus

- A mobile, intra-articular fibrocartilaginous crescentic plate.
- **Example:** Knee menisci.

2. Labrum

- A fixed, intra-articular fibrocartilaginous ring attached to the margin of an articular socket, serving to deepen the joint cavity.

- **Example:** Glenoid labrum of the shoulder joint.

3. Articular Disc

- A complete intra-articular fibrocartilaginous partition dividing the joint cavity.
- **Example:** Articular disc of the sternoclavicular joint.

Joint Mechanics

The primary physiological function of a synovial joint is to enable movement.

Each joint movement occurs around a specific axis and within a defined plane. The degree of joint mobility ranges from 0° to 180°.

Synovial joints facilitate several fundamental types of movement:

Flexion – Extension

- **Flexion:** Brings a body part forward or decreases the angle between segment parts.
- **Extension:** Moves a body part backward or increases the angle between segment parts.
- **Related Movements:** Dorsiflexion and Plantarflexion (talocrural joint).

Abduction – Adduction

- **Abduction:** Moves a body part away from the median plane.
- **Adduction:** Moves a body part toward the median plane.
- **Related Movements:** Radial and Ulnar deviation / inclination (radiocarpal joint).

External (Lateral) Rotation – Internal (Medial) Rotation

- **External Rotation:** A rotational movement directing a body part outward (medial to lateral).
- **Internal Rotation:** A rotational movement directing a body part inward (lateral to medial).
- **Related Movements:** Pronation and Supination (forearm).

Note: Circumduction

Circumduction is a composite, circular movement resulting from the sequential combination of four primary movements: adduction, abduction, flexion, and extension.